Project INSPIRE Design Document

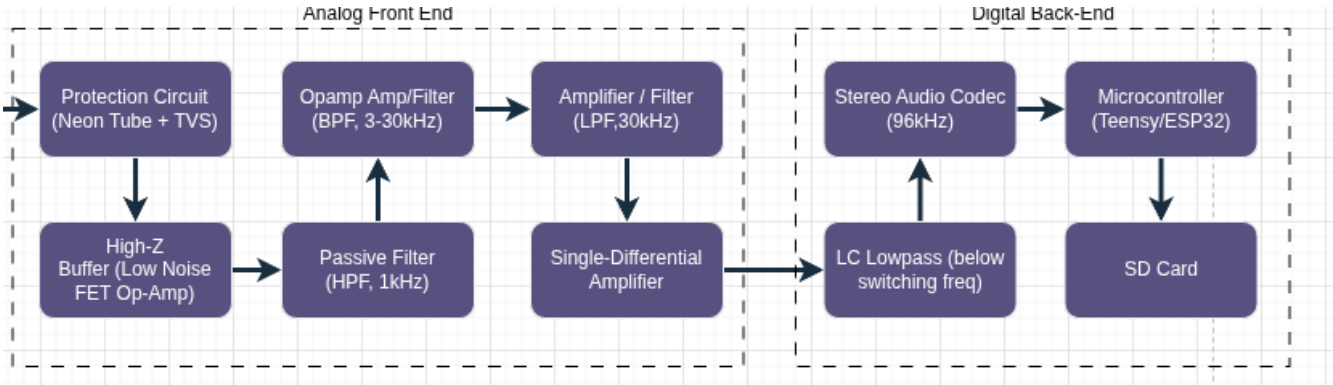
Project Part VI: Single-Differential Amplifier

Revision 1.0: Presented 10/9/2026

1) General Project Specifications

The purpose of the project is to build a receiver to capture natural radio atmospheric phenomena in the Very Low Frequency (VLF) range of 3-30kHz.

A block diagram of the project is provided below:



The project will be built as two Printed Circuit Boards (PCBs) inside shielded metal enclosures, and powered off batteries, in order to establish isolation and distance from power lines, which are very large emitters of interference.

In order to pick up these signals, a high resolution 24-bit audio Analog-Digital Converter (ADC) is required. As such, we can establish a dynamic range of approximately 100dB. This means that the analog stage should be prioritized to ensure linearity and prevent clipping.

2) Sub-circuit Specifications

In order to reject common-mode noise and ensure a low-loss connection between Analog and Digital boards, XLR audio cables are the best choice, as they can manage a 0.01dB/m attenuation. Because we are using XLR cables, we need to drive a differential signal between them, which requires a circuit called a Fully Differential Amplifier (FDA). This requires:

- Accurate Signal Conversion: Does the circuit convert the single-ended output to a balanced differential output (V_{in+}, V_{in-}), and does it do it across the measurement spectrum?
- Gain Stage: Does the block provide unity gain (+0dB) across the measurement spectrum, and does not have major gain ripple?
- Voltage Restrictions: Does it work with the 3.3V dual rail power supply system the analog signal chain will need to run on?

3) Expected Outputs

This sub-circuit requires a technical write-up and simulation covering the following parts:

- **Topology and Component Values:** A choice for a dual op-amp cross-coupled amplifier or a dedicated fully differential amplifier, as well as resistor values for these amplifiers
- **Component Selection:** Regardless of topology, a suitable operational amplifier must be chosen, ensuring minimization of Total Harmonic Distortion (THD), proper output voltage swing, sufficient GBW and Slew Rate, and a reasonable package type.
- **SPICE Simulation**: A SPICE simulation should be given to check the following:
  - Are the AC Sweeps showing a 180 degree phase across the measurement band, at unity gain (+0dB)?
  - Does the noise analysis show any major noise floor spikes or a non-linear behavior that would ruin the signal chain?

Optional Extra (Stretch Goals):

- Does the FDA support operation as a lowpass filter? If so, attempt to set a cutoff frequency at 48kHz to add another level of filtering before the ADC.